

Task 1:

When you go to a restaurant, it's polite to leave the waiter (/waitress) a tip. You've also got to pay sales tax on your meal, but not on the tip. For this task, we'll write a function that helps us calculate the total bill we'll pay. Your function should use MD's sales tax rate (6%) by default, but the user should be able to pass a different value if they want. Additionally, the function should default to a 15% tip, but again, you should be able to pass a different one if you want.

Task 2:

A few minutes ago, we stubbed out the `stddev` (standard deviation) and `mean` (arithmetic average) functions, but didn't implement them. Implement both of these functions, using the `mean` function as a helper function that is called by `stddev`. If you want a reminder on how to calculate standard deviation, you can see it [here](#).

You can test your `stddev` function with something like the following:

```
nums = [10, 7, 14, 3, 6, 21]

stddev_nums = stddev(nums)
```

The correct answer for the data above is 5.92.....

Task 3:

Write a function `primes_up_to_n(upper_bound)` that calculates and prints all prime numbers up to an upper bound specified. For this, you'll probably want to use a **helper function** that can determine if a single number is prime or not, which you can outsource part of the work to. Recall, we wrote this function previously.

Now, write another function, `first_n_primes(how_many)` that calculates and prints the first `n` primes. You should be able to *reuse* the helper function you wrote previously. This is a benefit of writing functions!

Careful – think about which type of loop (`for` vs `while`) is best suited for solving this part of the problem. It might not be the same as the one from the previous part.